

HECTOR GENARO PACHECO

Software Engineer | Backend • Cloud Infrastructure • DevOps

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PROFESSIONAL SUMMARY

Backend and cloud-focused software engineer with experience across standalone cloud architectures and collaborative team development in shared codebases. Hands-on experience building containerized REST & GraphQL microservices, relational PostgreSQL/Prisma schemas, AWS ECS Fargate deployments, zero-secret runtime configuration (AWS Secrets Manager/KMS), Jira-driven ticket workflows, Git/GitLab version control, and automated testing.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Bash/Shell	Backend & APIs: Node.js, Express, REST APIs, GraphQL, Apollo Server, FastAPI, Zod, RBAC
Cloud & Infra: AWS (ECS Fargate, ECR, RDS, Secrets Mgr, KMS, VPC, IAM), Vercel	Containers & OS: Docker, Multi-Stage Builds, Container Registries (ECR), Linux, systemd
Databases: PostgreSQL, Prisma ORM, Neon, Supabase, Relational Modeling, Indexing	DevOps & Testing: Git, GitLab, GitHub, Jira, Docker CLI, Vitest, Integration Testing, Build Verification
Architecture: Microservices, API Contract Design, Data Isolation, Deterministic Math	Security & Auth: JWT Authentication, HTTP-only Cookies, Secrets Injection, Least Privilege

FEATURED SOFTWARE & SYSTEMS ENGINEERING PROJECTS

CalTrack — *Industrial Calibration SaaS & API Platform* (Jun 2026 – Jul 2026) — [Node.js](#) • [Express](#) • [TypeScript](#) • [PostgreSQL](#) • [Prisma](#) • [Docker](#) • [AWS ECS Fargate](#) • [ECR](#) [LIVE APP](#) [GITHUB](#)

- Architected and deployed containerized Node/Express REST API on AWS ECS Fargate with Amazon RDS PostgreSQL in isolated VPC subnets.
- Built multi-stage Docker pipeline compiling shared TypeScript packages, generating Prisma client, running automated tests, and building linux/amd64 images.
- Configured Amazon ECR image hosting and injected runtime environment credentials securely using AWS Secrets Manager and AWS KMS.
- Enforced role-based access control (RBAC), database foreign-key constraints, and historical snapshot preservation for compliance auditing.

EstateOS — *Agentic Real Estate Investment Intelligence Platform* (Aug 2026 – Sep 2026) — [TypeScript](#) • [Node.js](#) • [Next.js 16](#) • [Claude API](#) • [PostgreSQL](#) • [Prisma ORM](#) • [Vitest](#) [CASE STUDY](#)

- Architected multi-agent orchestration service integrating Claude API research with deterministic underwriting engines and real-time SSE progress streaming.
- Engineered server-side evidence snapshot repositories in PostgreSQL/Prisma, enforcing source-authority tiers and cryptographic deduplication to isolate trusted data.
- Achieved 1,234 passing automated tests across 76 tracked Vitest suites, validating financial invariant math, agent workflows, and database repository transactions.

Procurement Tracking System — *Enterprise GraphQL Platform* (May 2026 – Jul 2026) — [Node.js](#) • [Express 5](#) • [GraphQL](#) • [Apollo Server 4](#) • [Prisma ORM 6](#) • [PostgreSQL](#) • [JWT](#) [LIVE APP](#) [GITHUB](#)

- Designed and deployed a type-safe GraphQL purchasing API on Render with PostgreSQL on Neon, modeling complex purchase-order and multi-stage receiving state transitions.
- Implemented Apollo Server 4 query/mutation resolvers with JWT authentication, HTTP-only session cookies, and multi-role RBAC resolver permission guards.

Additional Repositories & Deployments: **Interactive Photo Booth** — Linux systemd Daemon (Raspberry Pi 5/Python) [CASE STUDY](#) • **OC Water Features** — Production Web Platform (Next.js/Supabase/Vercel) [LIVE SITE](#) • **Pi Arcade OS** — Embedded Linux System (Python/GPIO) [SITE](#) [GITHUB](#)

TECHNICAL & INDUSTRIAL EXPERIENCE

Software Engineering Contributor — *Nonprofit serving underserved communities* Collaborative Contribution

- Collaborated on an existing nonprofit codebase using Jira-driven ticket workflows, Git/GitLab version control, API integration investigation, local environment configuration, and shared build and test workflows.

Instrumentation & Controls Technician — *Process Control & Industrial Systems* Field Engineering Experience

- Maintained high-availability industrial control infrastructure, diagnosing electrical loops, PLC I/O wiring, relays, and field instrumentation across 480VAC, 120VAC, and 24VDC systems with systematic root-cause isolation.

EDUCATION & PROFESSIONAL TRAINING

Western Governors University (WGU) — B.S. Software Engineering (In Progress, 2026–Present)

Palantir Technologies — Foundry & AIP Learning Credentials (May 2026)

Mirion Technologies — SU-890 iCAM Operations & Maintenance (March 2026)